

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z3,Z4,Z5

Generated by MacAnalog-Symbolix

October 17, 2025

Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z3 Z4 Z5: $\frac{Z_3 Z_4 Z_5 g_m - Z_3 Z_4}{2Z_3 Z_4 g_m + 2Z_3 Z_5 g_m + 2Z_3 + Z_4 Z_5 g_m + Z_4}$

$$H(z) = \frac{Z_3 Z_4 Z_5 g_m - Z_3 Z_4}{2Z_3 Z_4 g_m + 2Z_3 Z_5 g_m + 2Z_3 + Z_4 Z_5 g_m + Z_4}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(\infty, \infty, R_3, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 s + R_3 g_m}{2C_4 C_5 R_3 s^2 + g_m + s(2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{g_m}}{2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5}$
 wo: $\frac{\sqrt{2}\sqrt{g_m}}{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}}$
 bandwidth: $\frac{2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5}{2C_4 C_5 R_3}$
 K-LP: R_3
 K-HP: 0
 K-BP: $-\frac{C_5 R_3}{2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5}$
 Qz: None
 Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\infty, \infty, R_3, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 R_5 s + R_3 R_5 g_m - R_3}{2C_4 C_5 R_3 R_5 s^2 + 2R_3 g_m + R_5 g_m + s(2C_4 R_3 R_5 g_m + 2C_4 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{R_5}\sqrt{2R_3 g_m + R_5 g_m + 1}}{2C_4 R_3 R_5 g_m + 2C_4 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5}$
 wo: $\frac{\sqrt{2}\sqrt{2R_3 g_m + R_5 g_m + 1}}{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{R_5}}$
 bandwidth: $\frac{2C_4 R_3 R_5 g_m + 2C_4 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5}{2C_4 C_5 R_3 R_5}$
 K-LP: $\frac{R_3 R_5 g_m - R_3}{2R_3 g_m + R_5 g_m + 1}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_3 R_5}{2C_4 R_3 R_5 g_m + 2C_4 R_3 + 2C_5 R_3 R_5 g_m + C_5 R_5}$
 Qz: None
 Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(\infty, \infty, R_3, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_3 g_m + s (C_5 R_3 R_5 g_m - C_5 R_3)}{g_m + s^2 (2C_4 C_5 R_3 R_5 g_m + 2C_4 C_5 R_3) + s (2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5 R_5 g_m + C_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}R_5g_m\sqrt{\frac{g_m}{R_5g_m+1}} + \sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{\frac{g_m}{R_5g_m+1}}}{2C_4R_3g_m + 2C_5R_3g_m + C_5R_5g_m + C_5} \\ \text{wo: } & \sqrt{\frac{g_m}{2C_4C_5R_3R_5g_m + 2C_4C_5R_3}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{g_m}{2C_4C_5R_3R_5g_m + 2C_4C_5R_3}}(2C_4R_3g_m + 2C_5R_3g_m + C_5R_5g_m + C_5)}{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}R_5g_m\sqrt{\frac{g_m}{R_5g_m+1}} + \sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{\frac{g_m}{R_5g_m+1}}} \\ \text{K-LP: } & R_3 \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_5R_3R_5g_m - C_5R_3}{2C_4R_3g_m + 2C_5R_3g_m + C_5R_5g_m + C_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.4 X-INVALID-NUMER-4 $Z(s) = \left(\infty, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 R_4 s + R_3 R_4 g_m}{2C_4 C_5 R_3 R_4 s^2 + 2R_3 g_m + R_4 g_m + s (2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 + C_5 R_4)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}\sqrt{g_m}\sqrt{2R_3+R_4}}{2C_4R_3R_4g_m + 2C_5R_3R_4g_m + 2C_5R_3 + C_5R_4} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{2R_3g_m+R_4g_m}}{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}} \\ \text{bandwidth: } & \frac{\sqrt{2R_3g_m+R_4g_m}(2C_4R_3R_4g_m + 2C_5R_3R_4g_m + 2C_5R_3 + C_5R_4)}{2C_4C_5R_3R_4\sqrt{g_m}\sqrt{2R_3+R_4}} \\ \text{K-LP: } & \frac{R_3R_4}{2R_3+R_4} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5R_3R_4}{2C_4R_3R_4g_m + 2C_5R_3R_4g_m + 2C_5R_3 + C_5R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.5 X-INVALID-NUMER-5 $Z(s) = \left(\infty, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 R_4 R_5 s + R_3 R_4 R_5 g_m - R_3 R_4}{2C_4 C_5 R_3 R_4 R_5 s^2 + 2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s (2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}\sqrt{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}{2C_4R_3R_4R_5g_m + 2C_4R_3R_4 + 2C_5R_3R_4R_5g_m + 2C_5R_3R_5 + C_5R_4R_5} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}} \\ \text{bandwidth: } & \frac{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}}{2C_4R_3R_4R_5g_m + 2C_4R_3R_4 + 2C_5R_3R_4R_5g_m + 2C_5R_3R_5 + C_5R_4R_5} \\ \text{K-LP: } & \frac{R_3R_4R_5g_m - R_3R_4}{2R_3R_4g_m + 2R_3R_5g_m + 2R_3 + R_4R_5g_m + R_4} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5R_3R_4R_5}{2C_4R_3R_4R_5g_m + 2C_4R_3R_4 + 2C_5R_3R_4R_5g_m + 2C_5R_3R_5 + C_5R_4R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.6 X-INVALID-NUMER-6 $Z(s) = \left(\infty, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_3 R_4 g_m + s (C_5 R_3 R_4 R_5 g_m - C_5 R_3 R_4)}{2R_3 g_m + R_4 g_m + s^2 (2C_4 C_5 R_3 R_4 R_5 g_m + 2C_4 C_5 R_3 R_4) + s (2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 R_5 g_m + 2C_5 R_3 + C_5 R_4 R_5 g_m + C_5 R_4)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{g_m}{R_5g_m+1}}\sqrt{2R_3+R_4} + \sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{g_m}{R_5g_m+1}}\sqrt{2R_3+R_4}}{2C_4R_3R_4g_m + 2C_5R_3R_4g_m + 2C_5R_3R_5g_m + 2C_5R_3 + C_5R_4R_5g_m + C_5R_4}$$

wo: $\sqrt{\frac{2R_3g_m+R_4g_m}{2C_4C_5R_3R_4R_5g_m+2C_4C_5R_3R_4}}$
bandwidth: $\frac{\sqrt{\frac{2R_3g_m+R_4g_m}{2C_4C_5R_3R_4R_5g_m+2C_4C_5R_3R_4}}(2C_4R_3R_4g_m+2C_5R_3R_4g_m+2C_5R_3R_5g_m+2C_5R_3+C_5R_4R_5g_m+C_5R_4)}{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{gm}{R_5g_m+1}}\sqrt{2R_3+R_4}+\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{gm}{R_5g_m+1}}\sqrt{2R_3+R_4}}$
K-LP: $\frac{R_3R_4}{2R_3+R_4}$
K-HP: 0
K-BP: $\frac{C_5R_3R_4R_5g_m-C_5R_3R_4}{2C_4R_3R_4g_m+2C_5R_3R_4g_m+2C_5R_3R_5g_m+2C_5R_3+C_5R_4R_5g_m+C_5R_4}$
Qz: None
Wz: None

8.7 X-INVALID-NUMER-7 $Z(s) = \left(\infty, \infty, \frac{1}{C_3s}, R_4, \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{-C_5R_4s + R_4g_m}{C_3C_5R_4s^2 + 2g_m + s(C_3R_4g_m + 2C_5R_4g_m + 2C_5)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_3}\sqrt{C_5}\sqrt{R_4}\sqrt{g_m}}{C_3R_4g_m+2C_5R_4g_m+2C_5}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_3}\sqrt{C_5}\sqrt{R_4}}$
bandwidth: $\frac{C_3R_4g_m+2C_5R_4g_m+2C_5}{C_3C_5R_4}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $-\frac{C_5R_4}{C_3R_4g_m+2C_5R_4g_m+2C_5}$
Qz: None
Wz: None

8.8 X-INVALID-NUMER-8 $Z(s) = \left(\infty, \infty, \frac{1}{C_3s}, R_4, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_5R_4R_5s + R_4R_5g_m - R_4}{C_3C_5R_4R_5s^2 + 2R_4g_m + 2R_5g_m + s(C_3R_4R_5g_m + C_3R_4 + 2C_5R_4R_5g_m + 2C_5R_5) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_3}\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}\sqrt{R_4g_m+R_5g_m+1}}{C_3R_4R_5g_m+C_3R_4+2C_5R_4R_5g_m+2C_5R_5}$
wo: $\frac{\sqrt{2R_4g_m+2R_5g_m+2}}{\sqrt{C_3}\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2R_4g_m+2R_5g_m+2}(C_3R_4R_5g_m+C_3R_4+2C_5R_4R_5g_m+2C_5R_5)}{2C_3C_5R_4R_5\sqrt{R_4g_m+R_5g_m+1}}$
K-LP: $\frac{R_4R_5g_m-R_4}{2R_4g_m+2R_5g_m+2}$
K-HP: 0
K-BP: $-\frac{C_5R_4R_5}{C_3R_4R_5g_m+C_3R_4+2C_5R_4R_5g_m+2C_5R_5}$
Qz: None
Wz: None

8.9 X-INVALID-NUMER-9 $Z(s) = \left(\infty, \infty, \frac{1}{C_3s}, R_4, R_5 + \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{R_4g_m + s(C_5R_4R_5g_m - C_5R_4)}{2g_m + s^2(C_3C_5R_4R_5g_m + C_3C_5R_4) + s(C_3R_4g_m + 2C_5R_4g_m + 2C_5R_5g_m + 2C_5)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_3}\sqrt{C_5}\sqrt{R_4}R_5g_m\sqrt{\frac{gm}{R_5g_m+1}}+\sqrt{2}\sqrt{C_3}\sqrt{C_5}\sqrt{R_4}\sqrt{\frac{gm}{R_5g_m+1}}}{C_3R_4g_m+2C_5R_4g_m+2C_5R_5g_m+2C_5}$
wo: $\sqrt{2}\sqrt{\frac{gm}{C_3C_5R_4R_5g_m+C_3C_5R_4}}$
bandwidth: $\frac{\sqrt{2}\sqrt{\frac{gm}{C_3C_5R_4R_5g_m+C_3C_5R_4}}(C_3R_4g_m+2C_5R_4g_m+2C_5R_5g_m+2C_5)}{\sqrt{2}\sqrt{C_3}\sqrt{C_5}\sqrt{R_4}R_5g_m\sqrt{\frac{gm}{R_5g_m+1}}+\sqrt{2}\sqrt{C_3}\sqrt{C_5}\sqrt{R_4}\sqrt{\frac{gm}{R_5g_m+1}}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_5R_4R_5g_m-C_5R_4}{C_3R_4g_m+2C_5R_4g_m+2C_5R_5g_m+2C_5}$
Qz: None
Wz: None

8.10 X-INVALID-NUMER-10 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_5 s + R_5 g_m - 1}{2g_m + s^2 (C_3 C_5 R_5 + 2C_4 C_5 R_5) + s (C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_5}\sqrt{R_5}\sqrt{g_m}\sqrt{C_3+2C_4}}{C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m}$
 wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_3 C_5 R_5 + 2C_4 C_5 R_5}}$
 bandwidth: $\frac{C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m}{\sqrt{C_5}\sqrt{R_5}\sqrt{C_3 + 2C_4}\sqrt{C_3 C_5 R_5 + 2C_4 C_5 R_5}}$
 K-LP: $\frac{R_5 g_m - 1}{2g_m}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_5}{C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m}$
 Qz: None
 Wz: None

8.11 X-INVALID-NUMER-11 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_4 s + R_4 g_m}{2g_m + s^2 (C_3 C_5 R_4 + 2C_4 C_5 R_4) + s (C_3 R_4 g_m + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_5}\sqrt{R_4}\sqrt{g_m}\sqrt{C_3+2C_4}}{C_3 R_4 g_m + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5}$
 wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_3 C_5 R_4 + 2C_4 C_5 R_4}}$
 bandwidth: $\frac{C_3 R_4 g_m + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5}{\sqrt{C_5}\sqrt{R_4}\sqrt{C_3 + 2C_4}\sqrt{C_3 C_5 R_4 + 2C_4 C_5 R_4}}$
 K-LP: $\frac{R_4}{2}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_4}{C_3 R_4 g_m + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5}$
 Qz: None
 Wz: None

8.12 X-INVALID-NUMER-12 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_4 R_5 s + R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + s^2 (C_3 C_5 R_4 R_5 + 2C_4 C_5 R_4 R_5) + s (C_3 R_4 R_5 g_m + C_3 R_4 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}\sqrt{C_3+2C_4}\sqrt{R_4 g_m + R_5 g_m + 1}}{C_3 R_4 R_5 g_m + C_3 R_4 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5}$
 wo: $\frac{\sqrt{2R_4 g_m + 2R_5 g_m + 2}}{\sqrt{C_3 C_5 R_4 R_5 + 2C_4 C_5 R_4 R_5}}$
 bandwidth: $\frac{\sqrt{2}\sqrt{2R_4 g_m + 2R_5 g_m + 2}(C_3 R_4 R_5 g_m + C_3 R_4 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5)}{2\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}\sqrt{C_3 + 2C_4}\sqrt{C_3 C_5 R_4 R_5 + 2C_4 C_5 R_4 R_5}\sqrt{R_4 g_m + R_5 g_m + 1}}$
 K-LP: $\frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_4 R_5}{C_3 R_4 R_5 g_m + C_3 R_4 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5}$
 Qz: None
 Wz: None

8.13 X-INVALID-NUMER-13 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_4 g_m + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^2 (C_3 C_5 R_4 R_5 g_m + C_3 C_5 R_4 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_4) + s (C_3 R_4 g_m + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}C_3\sqrt{C_5}\sqrt{R_4}R_5g_m\sqrt{\frac{g_m}{C_3R_5g_m+C_3+2C_4R_5g_m+2C_4}}+\sqrt{2}C_3\sqrt{C_5}\sqrt{R_4}\sqrt{\frac{g_m}{C_3R_5g_m+C_3+2C_4R_5g_m+2C_4}}+2\sqrt{2}C_4\sqrt{C_5}\sqrt{R_4}R_5g_m\sqrt{\frac{g_m}{C_3R_5g_m+C_3+2C_4R_5g_m+2C_4}}+2\sqrt{2}C_4\sqrt{C_5}\sqrt{R_4}\sqrt{\frac{g_m}{C_3R_5g_m+C_3+2C_4R_5g_m+2C_4}}}{C_3R_4g_m+2C_4R_4g_m+2C_5R_4g_m+2C_5R_5g_m+2C_5} \\ \text{wo: } & \sqrt{2}\sqrt{\frac{g_m}{C_3C_5R_4R_5g_m+C_3C_5R_4+2C_4C_5R_4R_5g_m+2C_4C_5R_4}} \\ \text{bandwidth: } & \frac{\sqrt{2}\sqrt{\frac{g_m}{C_3C_5R_4R_5g_m+C_3C_5R_4+2C_4C_5R_4R_5g_m+2C_4C_5R_4}}(C_3R_4g_m+2C_4R_4g_m+2C_5R_4g_m+2C_5R_5g_m+2C_5)}{\sqrt{2}C_3\sqrt{C_5}\sqrt{R_4}R_5g_m\sqrt{\frac{g_m}{C_3R_5g_m+C_3+2C_4R_5g_m+2C_4}}+\sqrt{2}C_3\sqrt{C_5}\sqrt{R_4}\sqrt{\frac{g_m}{C_3R_5g_m+C_3+2C_4R_5g_m+2C_4}}+2\sqrt{2}C_4\sqrt{C_5}\sqrt{R_4}R_5g_m\sqrt{\frac{g_m}{C_3R_5g_m+C_3+2C_4R_5g_m+2C_4}}+2\sqrt{2}C_4\sqrt{C_5}\sqrt{R_4}\sqrt{\frac{g_m}{C_3R_5g_m+C_3+2C_4R_5g_m+2C_4}}} \\ \text{K-LP: } & \frac{R_4}{2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_5R_4R_5g_m-C_5R_4}{C_3R_4g_m+2C_4R_4g_m+2C_5R_4g_m+2C_5R_5g_m+2C_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.14 X-INVALID-NUMER-14 $Z(s) = \left(\infty, \infty, \frac{1}{C_3s}, R_4 + \frac{1}{C_4s}, R_5, \infty \right)$

$$H(s) = \frac{R_5g_m + s(C_4R_4R_5g_m - C_4R_4) - 1}{2g_m + s^2(C_3C_4R_4R_5g_m + C_3C_4R_4) + s(C_3R_5g_m + C_3 + 2C_4R_4g_m + 2C_4R_5g_m + 2C_4)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_3}\sqrt{C_4}\sqrt{R_4}R_5g_m\sqrt{\frac{g_m}{R_5g_m+1}}+\sqrt{2}\sqrt{C_3}\sqrt{C_4}\sqrt{R_4}\sqrt{\frac{g_m}{R_5g_m+1}}}{C_3R_5g_m+C_3+2C_4R_4g_m+2C_4R_5g_m+2C_4} \\ \text{wo: } & \sqrt{2}\sqrt{\frac{g_m}{C_3C_4R_4R_5g_m+C_3C_4R_4}} \\ \text{bandwidth: } & \frac{\sqrt{2}\sqrt{\frac{g_m}{C_3C_4R_4R_5g_m+C_3C_4R_4}}(C_3R_5g_m+C_3+2C_4R_4g_m+2C_4R_5g_m+2C_4)}{\sqrt{2}\sqrt{C_3}\sqrt{C_4}\sqrt{R_4}R_5g_m\sqrt{\frac{g_m}{R_5g_m+1}}+\sqrt{2}\sqrt{C_3}\sqrt{C_4}\sqrt{R_4}\sqrt{\frac{g_m}{R_5g_m+1}}} \\ \text{K-LP: } & \frac{R_5g_m-1}{2g_m} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_4R_4R_5g_m-C_4R_4}{C_3R_5g_m+C_3+2C_4R_4g_m+2C_4R_5g_m+2C_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.15 X-INVALID-NUMER-15 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3R_3s+1}, R_4, \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{-C_5R_3R_4s + R_3R_4g_m}{C_3C_5R_3R_4s^2 + 2R_3g_m + R_4g_m + s(C_3R_3R_4g_m + 2C_5R_3R_4g_m + 2C_5R_3 + C_5R_4)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_3}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}\sqrt{g_m}\sqrt{2R_3+R_4}}{C_3R_3R_4g_m+2C_5R_3R_4g_m+2C_5R_3+C_5R_4} \\ \text{wo: } & \frac{\sqrt{2R_3g_m+R_4g_m}}{\sqrt{C_3}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}} \\ \text{bandwidth: } & \frac{\sqrt{2R_3g_m+R_4g_m}(C_3R_3R_4g_m+2C_5R_3R_4g_m+2C_5R_3+C_5R_4)}{C_3C_5R_3R_4\sqrt{g_m}\sqrt{2R_3+R_4}} \\ \text{K-LP: } & \frac{R_3R_4}{2R_3+R_4} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5R_3R_4}{C_3R_3R_4g_m+2C_5R_3R_4g_m+2C_5R_3+C_5R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.16 X-INVALID-NUMER-16 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3R_3s+1}, R_4, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_5R_3R_4R_5s + R_3R_4R_5g_m - R_3R_4}{C_3C_5R_3R_4R_5s^2 + 2R_3R_4g_m + 2R_3R_5g_m + 2R_3 + R_4R_5g_m + R_4 + s(C_3R_3R_4R_5g_m + C_3R_3R_4 + 2C_5R_3R_4R_5g_m + 2C_5R_3R_5 + C_5R_4R_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_3}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}\sqrt{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}{C_3R_3R_4R_5g_m+C_3R_3R_4+2C_5R_3R_4R_5g_m+2C_5R_3R_5+C_5R_4R_5} \\ \text{wo: } & \frac{\sqrt{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}{\sqrt{C_3}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}\sqrt{R_5}} \\ \text{bandwidth: } & \frac{\sqrt{2R_3R_4R_5g_m+2R_3R_4+2C_5R_3R_4R_5g_m+2C_5R_3R_5+C_5R_4R_5}}{C_3C_5R_3R_4R_5} \\ \text{K-LP: } & \frac{R_3R_4R_5g_m-R_3R_4}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5R_3R_4R_5}{C_3R_3R_4R_5g_m+C_3R_3R_4+2C_5R_3R_4R_5g_m+2C_5R_3R_5+C_5R_4R_5} \end{aligned}$$

Qz: None
Wz: None

8.17 X-INVALID-NUMER-17 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_3 R_4 g_m + s (C_5 R_3 R_4 R_5 g_m - C_5 R_3 R_4)}{2 R_3 g_m + R_4 g_m + s^2 (C_3 C_5 R_3 R_4 R_5 g_m + C_3 C_5 R_3 R_4) + s (C_3 R_3 R_4 g_m + 2 C_5 R_3 R_4 g_m + 2 C_5 R_3 R_5 g_m + 2 C_5 R_3 + C_5 R_4 R_5 g_m + C_5 R_4)}$$

Parameters:

Q: $\frac{\sqrt{C_3} \sqrt{C_5} \sqrt{R_3} \sqrt{R_4} R_5 g_m \sqrt{\frac{g_m}{R_5 g_m + 1}} \sqrt{2 R_3 + R_4} + \sqrt{C_3} \sqrt{C_5} \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{R_5 g_m + 1}} \sqrt{2 R_3 + R_4}}{C_3 R_3 R_4 g_m + 2 C_5 R_3 R_4 g_m + 2 C_5 R_3 R_5 g_m + 2 C_5 R_3 + C_5 R_4 R_5 g_m + C_5 R_4}$
wo: $\sqrt{\frac{2 R_3 g_m + R_4 g_m}{C_3 C_5 R_3 R_4 R_5 g_m + C_3 C_5 R_3 R_4}}$
bandwidth: $\frac{\sqrt{\frac{2 R_3 g_m + R_4 g_m}{C_3 C_5 R_3 R_4 R_5 g_m + C_3 C_5 R_3 R_4}} (C_3 R_3 R_4 g_m + 2 C_5 R_3 R_4 g_m + 2 C_5 R_3 R_5 g_m + 2 C_5 R_3 + C_5 R_4 R_5 g_m + C_5 R_4)}{\sqrt{C_3} \sqrt{C_5} \sqrt{R_3} \sqrt{R_4} R_5 g_m \sqrt{\frac{g_m}{R_5 g_m + 1}} \sqrt{2 R_3 + R_4} + \sqrt{C_3} \sqrt{C_5} \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{R_5 g_m + 1}} \sqrt{2 R_3 + R_4}}$
K-LP: $\frac{R_3 R_4}{2 R_3 + R_4}$
K-HP: 0
K-BP: $\frac{C_5 R_3 R_4 R_5 g_m - C_5 R_3 R_4}{C_3 R_3 R_4 g_m + 2 C_5 R_3 R_4 g_m + 2 C_5 R_3 R_5 g_m + 2 C_5 R_3 + C_5 R_4 R_5 g_m + C_5 R_4}$
Qz: None
Wz: None

8.18 X-INVALID-NUMER-18 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 s + R_3 g_m}{g_m + s^2 (C_3 C_5 R_3 + 2 C_4 C_5 R_3) + s (C_3 R_3 g_m + 2 C_4 R_3 g_m + 2 C_5 R_3 g_m + C_5)}$$

Parameters:

Q: $\frac{\sqrt{C_5} \sqrt{R_3} \sqrt{g_m} \sqrt{C_3 + 2 C_4}}{C_3 R_3 g_m + 2 C_4 R_3 g_m + 2 C_5 R_3 g_m + C_5}$
wo: $\frac{\sqrt{g_m}}{\sqrt{C_3 C_5 R_3 + 2 C_4 C_5 R_3}}$
bandwidth: $\frac{C_3 R_3 g_m + 2 C_4 R_3 g_m + 2 C_5 R_3 g_m + C_5}{\sqrt{C_5} \sqrt{R_3} \sqrt{C_3 + 2 C_4} \sqrt{C_3 C_5 R_3 + 2 C_4 C_5 R_3}}$
K-LP: R_3
K-HP: 0
K-BP: $-\frac{C_5 R_3}{C_3 R_3 g_m + 2 C_4 R_3 g_m + 2 C_5 R_3 g_m + C_5}$
Qz: None
Wz: None

8.19 X-INVALID-NUMER-19 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 R_5 s + R_3 R_5 g_m - R_3}{2 R_3 g_m + R_5 g_m + s^2 (C_3 C_5 R_3 R_5 + 2 C_4 C_5 R_3 R_5) + s (C_3 R_3 R_5 g_m + C_3 R_3 + 2 C_4 R_3 R_5 g_m + 2 C_4 R_3 + 2 C_5 R_3 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_5} \sqrt{R_3} \sqrt{R_5} \sqrt{C_3 + 2 C_4} \sqrt{2 R_3 g_m + R_5 g_m + 1}}{C_3 R_3 R_5 g_m + C_3 R_3 + 2 C_4 R_3 R_5 g_m + 2 C_4 R_3 + 2 C_5 R_3 R_5 g_m + C_5 R_5}$
wo: $\frac{\sqrt{2 R_3 g_m + R_5 g_m + 1}}{\sqrt{C_3 C_5 R_3 R_5 + 2 C_4 C_5 R_3 R_5}}$
bandwidth: $\frac{C_3 R_3 R_5 g_m + C_3 R_3 + 2 C_4 R_3 R_5 g_m + 2 C_4 R_3 + 2 C_5 R_3 R_5 g_m + C_5 R_5}{\sqrt{C_5} \sqrt{R_3} \sqrt{R_5} \sqrt{C_3 + 2 C_4} \sqrt{C_3 C_5 R_3 R_5 + 2 C_4 C_5 R_3 R_5}}$
K-LP: $\frac{R_3 R_5 g_m - R_3}{2 R_3 g_m + R_5 g_m + 1}$
K-HP: 0
K-BP: $-\frac{C_5 R_3 R_5}{C_3 R_3 R_5 g_m + C_3 R_3 + 2 C_4 R_3 R_5 g_m + 2 C_4 R_3 + 2 C_5 R_3 R_5 g_m + C_5 R_5}$
Qz: None
Wz: None

8.20 X-INVALID-NUMER-20 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_3 g_m + s (C_5 R_3 R_5 g_m - C_5 R_3)}{g_m + s^2 (C_3 C_5 R_3 R_5 g_m + C_3 C_5 R_3 + 2C_4 C_5 R_3 R_5 g_m + 2C_4 C_5 R_3) + s (C_3 R_3 g_m + 2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5 R_5 g_m + C_5)}$$

Parameters:

Q: $\frac{C_3 \sqrt{C_5} \sqrt{R_3} R_5 g_m \sqrt{\frac{g_m}{C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4}} + C_3 \sqrt{C_5} \sqrt{R_3} \sqrt{\frac{g_m}{C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4}} + 2C_4 \sqrt{C_5} \sqrt{R_3} R_5 g_m \sqrt{\frac{g_m}{C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4}} + 2C_4 \sqrt{C_5} \sqrt{R_3} \sqrt{\frac{g_m}{C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4}}}{C_3 R_3 g_m + 2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5 R_5 g_m + C_5}$

wo: $\sqrt{\frac{g_m}{C_3 C_5 R_3 R_5 g_m + C_3 C_5 R_3 + 2C_4 C_5 R_3 R_5 g_m + 2C_4 C_5 R_3}}$

bandwidth: $\frac{\sqrt{\frac{g_m}{C_3 C_5 R_3 R_5 g_m + C_3 C_5 R_3 + 2C_4 C_5 R_3 R_5 g_m + 2C_4 C_5 R_3}} (C_3 R_3 g_m + 2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5 R_5 g_m + C_5)}{C_3 \sqrt{C_5} \sqrt{R_3} R_5 g_m \sqrt{\frac{g_m}{C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4}} + C_3 \sqrt{C_5} \sqrt{R_3} \sqrt{\frac{g_m}{C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4}} + 2C_4 \sqrt{C_5} \sqrt{R_3} R_5 g_m \sqrt{\frac{g_m}{C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4}} + 2C_4 \sqrt{C_5} \sqrt{R_3} \sqrt{\frac{g_m}{C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4}}}$

K-LP: R_3

K-HP: 0

K-BP: $\frac{C_5 R_3 R_5 g_m - C_5 R_3}{C_3 R_3 g_m + 2C_4 R_3 g_m + 2C_5 R_3 g_m + C_5 R_5 g_m + C_5}$

Qz: None

Wz: None

8.21 X-INVALID-NUMER-21 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 R_4 s + R_3 R_4 g_m}{2R_3 g_m + R_4 g_m + s^2 (C_3 C_5 R_3 R_4 + 2C_4 C_5 R_3 R_4) + s (C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 + C_5 R_4)}$$

Parameters:

Q: $\frac{\sqrt{C_5} \sqrt{R_3} \sqrt{R_4} \sqrt{g_m} \sqrt{C_3 + 2C_4} \sqrt{2R_3 + R_4}}{C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 + C_5 R_4}$

wo: $\frac{\sqrt{2R_3 g_m + R_4 g_m}}{\sqrt{C_3 C_5 R_3 R_4 + 2C_4 C_5 R_3 R_4}}$

bandwidth: $\frac{\sqrt{2R_3 g_m + R_4 g_m} (C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 + C_5 R_4)}{\sqrt{C_5} \sqrt{R_3} \sqrt{R_4} \sqrt{g_m} \sqrt{C_3 + 2C_4} \sqrt{2R_3 + R_4} \sqrt{C_3 C_5 R_3 R_4 + 2C_4 C_5 R_3 R_4}}$

K-LP: $\frac{R_3 R_4}{2R_3 + R_4}$

K-HP: 0

K-BP: $-\frac{C_5 R_3 R_4}{C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 + C_5 R_4}$

Qz: None

Wz: None

8.22 X-INVALID-NUMER-22 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_3 R_4 R_5 s + R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s^2 (C_3 C_5 R_3 R_4 R_5 + 2C_4 C_5 R_3 R_4 R_5) + s (C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_5} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{C_3 + 2C_4} \sqrt{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}}{C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5}$

wo: $\frac{\sqrt{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}}{\sqrt{C_3 C_5 R_3 R_4 R_5 + 2C_4 C_5 R_3 R_4 R_5}}$

bandwidth: $\frac{C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5}{\sqrt{C_5} \sqrt{R_3} \sqrt{R_4} \sqrt{R_5} \sqrt{C_3 + 2C_4} \sqrt{C_3 C_5 R_3 R_4 R_5 + 2C_4 C_5 R_3 R_4 R_5}}$

K-LP: $\frac{R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}$

K-HP: 0

K-BP: $-\frac{C_5 R_3 R_4 R_5}{C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4 + 2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5}$

Qz: None

Wz: None

8.23 X-INVALID-NUMER-23 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_3 R_4 g_m + s (C_5 R_3 R_4 R_5 g_m - C_5 R_3 R_4)}{2R_3 g_m + R_4 g_m + s^2 (C_3 C_5 R_3 R_4 R_5 g_m + C_3 C_5 R_3 R_4 + 2C_4 C_5 R_3 R_4 R_5 g_m + 2C_4 C_5 R_3 R_4) + s (C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 R_5 g_m + 2C_5 R_3 + C_5 R_4 R_5 g_m + C_5 R_4)}$$

Parameters:

Q: $\frac{C_3 \sqrt{C_5} \sqrt{R_3} \sqrt{R_4} R_5 g_m \sqrt{\frac{g_m}{C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4}} \sqrt{2R_3 + R_4} + C_3 \sqrt{C_5} \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4}} \sqrt{2R_3 + R_4} + 2C_4 \sqrt{C_5} \sqrt{R_3} \sqrt{R_4} R_5 g_m \sqrt{\frac{g_m}{C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4}} \sqrt{2R_3 + R_4} + 2C_4 \sqrt{C_5} \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4}} \sqrt{2R_3 + R_4}}{C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m + 2C_5 R_3 R_4 g_m + 2C_5 R_3 R_5 g_m + 2C_5 R_3 + C_5 R_4 R_5 g_m + C_5 R_4}$

wo: $\sqrt{\frac{2R_3g_m+R_4g_m}{C_3C_5R_3R_4R_5g_m+C_3C_5R_3R_4+2C_4C_5R_3R_4R_5g_m+2C_4C_5R_3R_4}}$

bandwidth: $\frac{\sqrt{\frac{2R_3g_m+R_4g_m}{C_3C_5R_3R_4R_5g_m+C_3C_5R_3R_4+2C_4C_5R_3R_4R_5g_m+2C_4C_5R_3R_4}}(C_3R_3R_4g_m+2C_4R_3R_4g_m+2C_5R_3R_4g_m+2C_5R_3R_5g_m+2C_5R_3+C_5R_4R_5g_m+C_5R_4)}{C_3\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{g_m}{C_3R_5g_m+C_3+2C_4R_5g_m+2C_4}}\sqrt{2R_3+R_4+C_3}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{g_m}{C_3R_5g_m+C_3+2C_4R_5g_m+2C_4}}\sqrt{2R_3+R_4+2C_4}\sqrt{C_5}\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{g_m}{C_3R_5g_m+C_3+2C_4R_5g_m+2C_4}}\sqrt{2R_3+R_4}}$

K-LP: $\frac{R_3R_4}{2R_3+R_4}$
K-HP: 0
K-BP: $\frac{C_5R_3R_4R_5g_m-C_5R_3R_4}{C_3R_3R_4g_m+2C_4R_3R_4g_m+2C_5R_3R_4g_m+2C_5R_3R_5g_m+2C_5R_3+C_5R_4R_5g_m+C_5R_4}$
Qz: None
Wz: None

8.24 X-INVALID-NUMER-24 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3R_3s+1}, R_4 + \frac{1}{C_4s}, R_5, \infty \right)$

$$H(s) = \frac{R_3R_5g_m - R_3 + s(C_4R_3R_4R_5g_m - C_4R_3R_4)}{2R_3g_m + R_5g_m + s^2(C_3C_4R_3R_4R_5g_m + C_3C_4R_3R_4) + s(C_3R_3R_5g_m + C_3R_3 + 2C_4R_3R_4g_m + 2C_4R_3R_5g_m + 2C_4R_3 + C_4R_4R_5g_m + C_4R_4) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{2R_3g_m}{R_5g_m+1}+\frac{R_5g_m}{R_5g_m+1}+\frac{1}{R_5g_m+1}}+\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{2R_3g_m}{R_5g_m+1}+\frac{R_5g_m}{R_5g_m+1}+\frac{1}{R_5g_m+1}}}{C_3R_3R_5g_m+C_3R_3+2C_4R_3R_4g_m+2C_4R_3R_5g_m+2C_4R_3+C_4R_4R_5g_m+C_4R_4}$

wo: $\sqrt{\frac{2R_3g_m+R_5g_m+1}{C_3C_4R_3R_4R_5g_m+C_3C_4R_3R_4}}$

bandwidth: $\frac{\sqrt{\frac{2R_3g_m+R_5g_m+1}{C_3C_4R_3R_4R_5g_m+C_3C_4R_3R_4}}(C_3R_3R_5g_m+C_3R_3+2C_4R_3R_4g_m+2C_4R_3R_5g_m+2C_4R_3+C_4R_4R_5g_m+C_4R_4)}{\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{2R_3g_m}{R_5g_m+1}+\frac{R_5g_m}{R_5g_m+1}+\frac{1}{R_5g_m+1}}+\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{2R_3g_m}{R_5g_m+1}+\frac{R_5g_m}{R_5g_m+1}+\frac{1}{R_5g_m+1}}}$

K-LP: $\frac{R_3R_5g_m-R_3}{2R_3g_m+R_5g_m+1}$
K-HP: 0
K-BP: $\frac{C_4R_3R_4R_5g_m-C_4R_3R_4}{C_3R_3R_5g_m+C_3R_3+2C_4R_3R_4g_m+2C_4R_3R_5g_m+2C_4R_3+C_4R_4R_5g_m+C_4R_4}$
Qz: None
Wz: None

8.25 X-INVALID-NUMER-25 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3s}, \frac{1}{C_4s}, R_5, \infty \right)$

$$H(s) = \frac{R_5g_m + s(C_3R_3R_5g_m - C_3R_3) - 1}{2g_m + s^2(2C_3C_4R_3R_5g_m + 2C_3C_4R_3) + s(2C_3R_3g_m + C_3R_5g_m + C_3 + 2C_4R_5g_m + 2C_4)}$$

Parameters:

Q: $\frac{2\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}R_5g_m\sqrt{\frac{g_m}{R_5g_m+1}}+2\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{\frac{g_m}{R_5g_m+1}}}{2C_3R_3g_m+C_3R_5g_m+C_3+2C_4R_5g_m+2C_4}$

wo: $\sqrt{\frac{g_m}{C_3C_4R_3R_5g_m+C_3C_4R_3}}$

bandwidth: $\frac{\sqrt{\frac{g_m}{C_3C_4R_3R_5g_m+C_3C_4R_3}}(2C_3R_3g_m+C_3R_5g_m+C_3+2C_4R_5g_m+2C_4)}{2\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}R_5g_m\sqrt{\frac{g_m}{R_5g_m+1}}+2\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{\frac{g_m}{R_5g_m+1}}}$

K-LP: $\frac{R_5g_m-1}{2g_m}$
K-HP: 0
K-BP: $\frac{C_3R_3R_5g_m-C_3R_3}{2C_3R_3g_m+C_3R_5g_m+C_3+2C_4R_5g_m+2C_4}$
Qz: None
Wz: None

8.26 X-INVALID-NUMER-26 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3s}, \frac{R_4}{C_4R_4s+1}, R_5, \infty \right)$

$$H(s) = \frac{R_4R_5g_m - R_4 + s(C_3R_3R_4R_5g_m - C_3R_3R_4)}{2R_4g_m + 2R_5g_m + s^2(2C_3C_4R_3R_4R_5g_m + 2C_3C_4R_3R_4) + s(2C_3R_3R_4g_m + 2C_3R_3R_5g_m + 2C_3R_3 + C_3R_4R_5g_m + C_3R_4 + 2C_4R_4R_5g_m + 2C_4R_4) + 2}$$

Parameters:

Q: $\frac{2\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{R_4g_m}{R_5g_m+1}+\frac{R_5g_m}{R_5g_m+1}+\frac{1}{R_5g_m+1}}+2\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{R_4g_m}{R_5g_m+1}+\frac{R_5g_m}{R_5g_m+1}+\frac{1}{R_5g_m+1}}}{2C_3R_3R_4g_m+2C_3R_3R_5g_m+2C_3R_3+C_3R_4R_5g_m+C_3R_4+2C_4R_4R_5g_m+2C_4R_4}$

wo: $\sqrt{\frac{R_4g_m+R_5g_m+1}{C_3C_4R_3R_4R_5g_m+C_3C_4R_3R_4}}$

bandwidth: $\frac{\sqrt{\frac{R_4g_m+R_5g_m+1}{C_3C_4R_3R_4R_5g_m+C_3C_4R_3R_4}}(2C_3R_3R_4g_m+2C_3R_3R_5g_m+2C_3R_3+C_3R_4R_5g_m+C_3R_4+2C_4R_4R_5g_m+2C_4R_4)}{2\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}R_5g_m\sqrt{\frac{R_4g_m}{R_5g_m+1}+\frac{R_5g_m}{R_5g_m+1}+\frac{1}{R_5g_m+1}}+2\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{R_4g_m}{R_5g_m+1}+\frac{R_5g_m}{R_5g_m+1}+\frac{1}{R_5g_m+1}}}$

K-LP: $\frac{R_4R_5g_m-R_4}{2R_4g_m+2R_5g_m+2}$

K-HP: 0
K-BP: $\frac{C_3 R_3 R_4 R_5 g_m - C_3 R_3 R_4}{2C_3 R_3 R_4 g_m + 2C_3 R_3 R_5 g_m + 2C_3 R_3 + C_3 R_4 R_5 g_m + C_3 R_4 + 2C_4 R_4 R_5 g_m + 2C_4 R_4}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (\infty, \infty, R_3, R_4, R_5, \infty)$

$$H(s) = \frac{R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(\infty, \infty, R_3, R_4, \frac{1}{C_5 s}, \infty\right)$

$$H(s) = \frac{-C_5 R_3 R_4 s + R_3 R_4 g_m}{2R_3 g_m + R_4 g_m + s(2C_5 R_3 R_4 g_m + 2C_5 R_3 + C_5 R_4)}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(\infty, \infty, R_3, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty\right)$

$$H(s) = \frac{-C_5 R_3 R_4 R_5 s + R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s(2C_5 R_3 R_4 R_5 g_m + 2C_5 R_3 R_5 + C_5 R_4 R_5)}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(\infty, \infty, R_3, R_4, R_5 + \frac{1}{C_5 s}, \infty\right)$

$$H(s) = \frac{R_3 R_4 g_m + s(C_5 R_3 R_4 R_5 g_m - C_5 R_3 R_4)}{2R_3 g_m + R_4 g_m + s(2C_5 R_3 R_4 g_m + 2C_5 R_3 R_5 g_m + 2C_5 R_3 + C_5 R_4 R_5 g_m + C_5 R_4)}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(\infty, \infty, R_3, \frac{1}{C_4 s}, R_5, \infty\right)$

$$H(s) = \frac{R_3 R_5 g_m - R_3}{2R_3 g_m + R_5 g_m + s(2C_4 R_3 R_5 g_m + 2C_4 R_3) + 1}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(\infty, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty\right)$

$$H(s) = \frac{R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s(2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(\infty, \infty, R_3, R_4 + \frac{1}{C_4 s}, R_5, \infty\right)$

$$H(s) = \frac{R_3 R_5 g_m - R_3 + s(C_4 R_3 R_4 R_5 g_m - C_4 R_3 R_4)}{2R_3 g_m + R_5 g_m + s(2C_4 R_3 R_4 g_m + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + C_4 R_4 R_5 g_m + C_4 R_4) + 1}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, R_4, R_5, \infty\right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + s(C_3 R_4 R_5 g_m + C_3 R_4) + 2}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, R_5, \infty \right)$$

$$H(s) = \frac{R_5 g_m - 1}{2g_m + s(C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4)}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 s + g_m}{s^2(C_3 C_5 + 2C_4 C_5) + s(C_3 g_m + 2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{g_m + s(C_5 R_5 g_m - C_5)}{s^2(C_3 C_5 R_5 g_m + C_3 C_5 + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s(C_3 g_m + 2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + s(C_3 R_4 R_5 g_m + C_3 R_4 + 2C_4 R_4 R_5 g_m + 2C_4 R_4) + 2}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_4 s^2 + g_m + s(C_4 R_4 g_m - C_5)}{C_3 C_4 C_5 R_4 s^3 + s^2(C_3 C_4 R_4 g_m + C_3 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5) + s(C_3 g_m + 2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_4 R_5 s^2 + R_5 g_m + s(C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5) - 1}{C_3 C_4 C_5 R_4 R_5 s^3 + 2g_m + s^2(C_3 C_4 R_4 R_5 g_m + C_3 C_4 R_4 + C_3 C_5 R_5 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_5) + s(C_3 R_5 g_m + C_3 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m)}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{g_m + s^2(C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4) + s(C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^3(C_3 C_4 C_5 R_4 R_5 g_m + C_3 C_4 C_5 R_4) + s^2(C_3 C_4 R_4 g_m + C_3 C_5 R_5 g_m + C_3 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s(C_3 g_m + 2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, R_5, \infty \right)$$

$$H(s) = \frac{R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s(C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, R_5, \infty \right)$$

$$H(s) = \frac{R_3 R_5 g_m - R_3}{2R_3 g_m + R_5 g_m + s(C_3 R_3 R_5 g_m + C_3 R_3 + 2C_4 R_3 R_5 g_m + 2C_4 R_3) + 1}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$$

$$H(s) = \frac{R_3 R_4 R_5 g_m - R_3 R_4}{2R_3 R_4 g_m + 2R_3 R_5 g_m + 2R_3 + R_4 R_5 g_m + R_4 + s(C_3 R_3 R_4 R_5 g_m + C_3 R_3 R_4 + 2C_4 R_3 R_4 R_5 g_m + 2C_4 R_3 R_4)}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_3 R_4 s^2 + R_3 g_m + s (C_4 R_3 R_4 g_m - C_5 R_3)}{C_3 C_4 C_5 R_3 R_4 s^3 + g_m + s^2 (C_3 C_4 R_3 R_4 g_m + C_3 C_5 R_3 + 2C_4 C_5 R_3 R_4 g_m + 2C_4 C_5 R_3 + C_4 C_5 R_4) + s (C_3 R_3 g_m + 2C_4 R_3 g_m + C_4 R_4 g_m + 2C_5 R_3 g_m + C_5)}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_3 R_4 R_5 s^2 + R_3 R_5 g_m - R_3 + s (C_4 R_3 R_4 R_5 g_m - C_4 R_3 R_4 - C_5 R_3 R_5)}{C_3 C_4 C_5 R_3 R_4 R_5 s^3 + 2R_3 g_m + R_5 g_m + s^2 (C_3 C_4 R_3 R_4 R_5 g_m + C_3 C_4 R_3 R_4 + C_3 C_5 R_3 R_5 + 2C_4 C_5 R_3 R_4 R_5 g_m + 2C_4 C_5 R_3 R_5 + C_4 C_5 R_4 R_5) + s (C_3 R_3 R_5 g_m + C_3 R_3 + 2C_4 R_3 R_4 g_m + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + C_4 R_4 R_5 g_m + C_4 R_4 + 2C_5 R_3 R_5 g_m + C_5 R_5) + 1}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_3 g_m + s^2 (C_4 C_5 R_3 R_4 R_5 g_m - C_4 C_5 R_3 R_4) + s (C_4 R_3 R_4 g_m + C_5 R_3 R_5 g_m - C_5 R_3)}{g_m + s^3 (C_3 C_4 C_5 R_3 R_4 R_5 g_m + C_3 C_4 C_5 R_3 R_4) + s^2 (C_3 C_4 R_3 R_4 g_m + C_3 C_5 R_3 R_5 g_m + C_3 C_5 R_3 + 2C_4 C_5 R_3 R_4 g_m + 2C_4 C_5 R_3 R_5 g_m + 2C_4 C_5 R_3 + C_4 C_5 R_4 R_5 g_m + C_4 C_5 R_4) + s (C_3 R_3 g_m + 2C_4 R_3 g_m + C_4 R_4 g_m + 2C_5 R_3 g_m + C_5 R_5 g_m + C_5)}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, R_4, R_5, \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s (C_3 R_3 R_4 R_5 g_m - C_3 R_3 R_4)}{2R_4 g_m + 2R_5 g_m + s (2C_3 R_3 R_4 g_m + 2C_3 R_3 R_5 g_m + 2C_3 R_3 + C_3 R_4 R_5 g_m + C_3 R_4) + 2}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_3 s^2 + g_m + s (C_3 R_3 g_m - C_5)}{2C_3 C_4 C_5 R_3 s^3 + s^2 (2C_3 C_4 R_3 g_m + 2C_3 C_5 R_3 g_m + C_3 C_5 + 2C_4 C_5) + s (C_3 g_m + 2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_3 R_5 s^2 + R_5 g_m + s (C_3 R_3 R_5 g_m - C_3 R_3 - C_5 R_5) - 1}{2C_3 C_4 C_5 R_3 R_5 s^3 + 2g_m + s^2 (2C_3 C_4 R_3 R_5 g_m + 2C_3 C_4 R_3 + 2C_3 C_5 R_3 R_5 g_m + C_3 C_5 R_5 + 2C_4 C_5 R_5) + s (2C_3 R_3 g_m + C_3 R_5 g_m + C_3 + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{g_m + s^2 (C_3 C_5 R_3 R_5 g_m - C_3 C_5 R_3) + s (C_3 R_3 g_m + C_5 R_5 g_m - C_5)}{s^3 (2C_3 C_4 C_5 R_3 R_5 g_m + 2C_3 C_4 C_5 R_3) + s^2 (2C_3 C_4 R_3 g_m + 2C_3 C_5 R_3 g_m + C_3 C_5 R_5 g_m + C_3 C_5 + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (C_3 g_m + 2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_3 R_4 s^2 + R_4 g_m + s (C_3 R_3 R_4 g_m - C_5 R_4)}{2C_3 C_4 C_5 R_3 R_4 s^3 + 2g_m + s^2 (2C_3 C_4 R_3 R_4 g_m + 2C_3 C_5 R_3 R_4 g_m + 2C_3 C_5 R_3 + C_3 C_5 R_4 + 2C_4 C_5 R_4) + s (2C_3 R_3 g_m + C_3 R_4 g_m + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5)}$$

$$\mathbf{9.27 \quad X-INVALID-ORDER-27} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_3 C_5 R_3 R_4 R_5 s^2 + R_4 R_5 g_m - R_4 + s (C_3 R_3 R_4 R_5 g_m - C_3 R_3 R_4 - C_5 R_4 R_5)}{2C_3 C_4 C_5 R_3 R_4 R_5 s^3 + 2R_4 g_m + 2R_5 g_m + s^2 (2C_3 C_4 R_3 R_4 R_5 g_m + 2C_3 C_4 R_3 R_4 + 2C_3 C_5 R_3 R_4 R_5 g_m + 2C_3 C_5 R_3 R_5 + C_3 C_5 R_4 R_5 + 2C_4 C_5 R_4 R_5) + s (2C_3 R_3 R_4 g_m + 2C_3 R_3 R_5 g_m + 2C_3 R_3 + C_3 R_4 R_5 g_m + C_3 R_4 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5) + 1}$$

$$\mathbf{9.28 \quad X-INVALID-ORDER-28} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_4 g_m + s^2 (C_3 C_5 R_3 R_4 R_5 g_m - C_3 C_5 R_3 R_4) + s (C_3 R_3 R_4 g_m + C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^3 (2C_3 C_4 C_5 R_3 R_4 R_5 g_m + 2C_3 C_4 C_5 R_3 R_4) + s^2 (2C_3 C_4 R_3 R_4 g_m + 2C_3 C_5 R_3 R_4 g_m + 2C_3 C_5 R_3 R_5 g_m + 2C_3 C_5 R_3 + C_3 C_5 R_4 R_5 g_m + C_3 C_5 R_4 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_4) + s (2C_3 R_3 g_m + C_3 R_4 g_m + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

9.29 X-INVALID-ORDER-29 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_3 C_4 C_5 R_3 R_4 s^3 + g_m + s^2 (C_3 C_4 R_3 R_4 g_m - C_3 C_5 R_3 - C_4 C_5 R_4) + s (C_3 R_3 g_m + C_4 R_4 g_m - C_5)}{s^3 (2C_3 C_4 C_5 R_3 R_4 g_m + 2C_3 C_4 C_5 R_3 + C_3 C_4 C_5 R_4) + s^2 (2C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m + 2C_3 C_5 R_3 g_m + C_3 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5) + s (C_3 g_m + 2C_4 g_m + 2C_5 g_m)}$$

9.30 X-INVALID-ORDER-30 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_3 C_4 C_5 R_3 R_4 R_5 s^3 + R_5 g_m + s^2 (C_3 C_4 R_3 R_4 R_5 g_m - C_3 C_4 R_3 R_4 - C_3 C_5 R_3 R_5 - C_4 C_5 R_4 R_5) + s (C_3 R_3 R_5 g_m - C_3 R_3 + C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5) - 1}{2g_m + s^3 (2C_3 C_4 C_5 R_3 R_4 R_5 g_m + 2C_3 C_4 C_5 R_3 R_5 + C_3 C_4 C_5 R_4 R_5) + s^2 (2C_3 C_4 R_3 R_4 g_m + 2C_3 C_4 R_3 R_5 g_m + 2C_3 C_4 R_3 + C_3 C_4 R_4 R_5 g_m + C_3 C_4 R_4 + 2C_3 C_5 R_3 R_5 g_m + C_3 C_5 R_5 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_5) + s (2C_3 R_3 g_m + C_3 R_5 g_m + C_3 + 2C_4 R_4 g_m + 2C_4 R_5 g_m)}$$

9.31 X-INVALID-ORDER-31 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{g_m + s^3 (C_3 C_4 C_5 R_3 R_4 R_5 g_m - C_3 C_4 C_5 R_3 R_4) + s^2 (C_3 C_4 R_3 R_4 g_m + C_3 C_5 R_3 R_5 g_m - C_3 C_5 R_3 + C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4) + s (C_3 R_3 g_m + C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^3 (2C_3 C_4 C_5 R_3 R_4 g_m + 2C_3 C_4 C_5 R_3 R_5 g_m + 2C_3 C_4 C_5 R_3 + C_3 C_4 C_5 R_4 R_5 g_m + C_3 C_4 C_5 R_4) + s^2 (2C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m + 2C_3 C_5 R_3 g_m + C_3 C_5 R_5 g_m + C_3 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (C_3 g_m + 2C_4 g_m + 2C_5 g_m)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(\infty, \infty, R_3, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 R_3 R_4 s^2 + R_3 g_m + s (C_4 R_3 R_4 g_m - C_5 R_3)}{g_m + s^2 (2C_4 C_5 R_3 R_4 g_m + 2C_4 C_5 R_3 + C_4 C_5 R_4) + s (2C_4 R_3 g_m + C_4 R_4 g_m + 2C_5 R_3 g_m + C_5)}$$

Parameters:

Q: $\frac{2\sqrt{C_4}\sqrt{C_5}R_3R_4g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3+R_4}}+2\sqrt{C_4}\sqrt{C_5}R_3\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3+R_4}}+\sqrt{C_4}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3+R_4}}}{2C_4R_3g_m+C_4R_4g_m+2C_5R_3g_m+C_5}$

wo: $\sqrt{\frac{g_m}{2C_4C_5R_3R_4g_m+2C_4C_5R_3+C_4C_5R_4}}$

bandwidth: $\frac{\sqrt{\frac{g_m}{2C_4C_5R_3R_4g_m+2C_4C_5R_3+C_4C_5R_4}}(2C_4R_3g_m+C_4R_4g_m+2C_5R_3g_m+C_5)}{2\sqrt{C_4}\sqrt{C_5}R_3R_4g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3+R_4}}+2\sqrt{C_4}\sqrt{C_5}R_3\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3+R_4}}+\sqrt{C_4}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3+R_4}}}$

K-LP: R_3

K-HP: $-\frac{R_3R_4}{2R_3R_4g_m+2R_3+R_4}$

K-BP: $\frac{C_4R_3R_4g_m-C_5R_3}{2C_4R_3g_m+C_4R_4g_m+2C_5R_3g_m+C_5}$

Qz: None

Wz: $\frac{\sqrt{-g_m}}{\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}}$

10.2 X-INVALID-WZ-2 $Z(s) = \left(\infty, \infty, R_3, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 R_3 R_4 R_5 s^2 + R_3 R_5 g_m - R_3 + s (C_4 R_3 R_4 R_5 g_m - C_4 R_3 R_4 - C_5 R_3 R_5)}{2R_3 g_m + R_5 g_m + s^2 (2C_4 C_5 R_3 R_4 R_5 g_m + 2C_4 C_5 R_3 R_5 + C_4 C_5 R_4 R_5) + s (2C_4 R_3 R_4 g_m + 2C_4 R_3 R_5 g_m + 2C_4 R_3 + C_4 R_4 R_5 g_m + C_4 R_4 + 2C_5 R_3 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{2\sqrt{C_4}\sqrt{C_5}R_3R_4\sqrt{R_5}g_m\sqrt{\frac{2R_3g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{R_5g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{1}{2R_3R_4g_m+2R_3+R_4}}+2\sqrt{C_4}\sqrt{C_5}R_3\sqrt{R_5}\sqrt{\frac{2R_3g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{R_5g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{1}{2R_3R_4g_m+2R_3+R_4}}+\sqrt{C_4}\sqrt{C_5}R_4\sqrt{R_5}\sqrt{\frac{2R_3g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{R_5g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{1}{2R_3R_4g_m+2R_3+R_4}}}{2C_4R_3R_4g_m+2C_4R_3R_5g_m+2C_4R_3+C_4R_4R_5g_m+C_4R_4+2C_5R_3R_5g_m+C_5R_5}$

wo: $\sqrt{\frac{2R_3g_m+R_5g_m+1}{2C_4C_5R_3R_4R_5g_m+2C_4C_5R_3R_5+C_4C_5R_4R_5}}$

bandwidth: $\frac{\sqrt{\frac{2R_3g_m+R_5g_m+1}{2C_4C_5R_3R_4R_5g_m+2C_4C_5R_3R_5+C_4C_5R_4R_5}}(2C_4R_3R_4g_m+2C_4R_3R_5g_m+2C_4R_3+C_4R_4R_5g_m+C_4R_4+2C_5R_3R_5g_m+C_5R_5)}{2\sqrt{C_4}\sqrt{C_5}R_3R_4\sqrt{R_5}g_m\sqrt{\frac{2R_3g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{R_5g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{1}{2R_3R_4g_m+2R_3+R_4}}+2\sqrt{C_4}\sqrt{C_5}R_3\sqrt{R_5}\sqrt{\frac{2R_3g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{R_5g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{1}{2R_3R_4g_m+2R_3+R_4}}+\sqrt{C_4}\sqrt{C_5}R_4\sqrt{R_5}\sqrt{\frac{2R_3g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{R_5g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{1}{2R_3R_4g_m+2R_3+R_4}}}$

K-LP: $\frac{R_3R_5g_m-R_3}{2R_3g_m+R_5g_m+1}$

K-HP: $-\frac{R_3R_4}{2R_3R_4g_m+2R_3+R_4}$

K-BP: $\frac{C_4R_3R_4R_5g_m-C_4R_3R_4-C_5R_3R_5}{2C_4R_3R_4g_m+2C_4R_3R_5g_m+2C_4R_3+C_4R_4R_5g_m+C_4R_4+2C_5R_3R_5g_m+C_5R_5}$

Qz: None

Wz: $\frac{\sqrt{-R_5g_m+1}}{\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}}$

10.3 X-INVALID-WZ-3 $Z(s) = \left(\infty, \infty, R_3, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_3 g_m + s^2 (C_4 C_5 R_3 R_4 R_5 g_m - C_4 C_5 R_3 R_4) + s (C_4 R_3 R_4 g_m + C_5 R_3 R_5 g_m - C_5 R_3)}{g_m + s^2 (2C_4 C_5 R_3 R_4 g_m + 2C_4 C_5 R_3 R_5 g_m + 2C_4 C_5 R_3 + C_4 C_5 R_4 R_5 g_m + C_4 C_5 R_4) + s (2C_4 R_3 g_m + C_4 R_4 g_m + 2C_5 R_3 g_m + C_5 R_5 g_m + C_5)}$$

Parameters:

Q: $\frac{2\sqrt{C_4}\sqrt{C_5}R_3R_4g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+2\sqrt{C_4}\sqrt{C_5}R_3R_5g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+2\sqrt{C_4}\sqrt{C_5}R_3\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+\sqrt{C_4}\sqrt{C_5}R_4R_5g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+\sqrt{C_4}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}}{2C_4R_3g_m+C_4R_4g_m+2C_5R_3g_m+C_5R_5g_m+C_5}$

wo: $\sqrt{\frac{g_m}{2C_4C_5R_3R_4g_m+2C_4C_5R_3R_5g_m+2C_4C_5R_3+C_4C_5R_4R_5g_m+C_4C_5R_4}}$

bandwidth: $\frac{\sqrt{\frac{g_m}{2C_4C_5R_3R_4g_m+2C_4C_5R_3R_5g_m+2C_4C_5R_3+C_4C_5R_4R_5g_m+C_4C_5R_4}}(2C_4R_3g_m+C_4R_4g_m+2C_5R_3g_m+C_5R_5g_m+C_5)}{2\sqrt{C_4}\sqrt{C_5}R_3R_4g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+2\sqrt{C_4}\sqrt{C_5}R_3R_5g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+2\sqrt{C_4}\sqrt{C_5}R_3\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+\sqrt{C_4}\sqrt{C_5}R_4R_5g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+\sqrt{C_4}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}}$

K-LP: R_3

K-HP: $\frac{R_3R_4R_5g_m-R_3R_4}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}$

K-BP: $\frac{C_4R_3R_4g_m+C_5R_3R_5g_m-C_5R_3}{2C_4R_3g_m+C_4R_4g_m+2C_5R_3g_m+C_5R_5g_m+C_5}$

Qz: None

Wz: $\sqrt{\frac{g_m}{C_4C_5R_4R_5g_m-C_4C_5R_4}}$

10.4 X-INVALID-WZ-4 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_3C_5R_3R_4s^2 + R_4g_m + s (C_3R_3R_4g_m - C_5R_4)}{2g_m + s^2 (2C_3C_5R_3R_4g_m + 2C_3C_5R_3 + C_3C_5R_4) + s (2C_3R_3g_m + C_3R_4g_m + 2C_5R_4g_m + 2C_5)}$$

Parameters:

Q: $\frac{2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_3R_4g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3+R_4}}+2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_3\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3+R_4}}+2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3+R_4}}}{2C_3R_3g_m+C_3R_4g_m+2C_5R_4g_m+2C_5}$

wo: $\sqrt{2}\sqrt{\frac{g_m}{2C_3C_5R_3R_4g_m+2C_3C_5R_3+C_3C_5R_4}}$

bandwidth: $\frac{\sqrt{2}\sqrt{\frac{g_m}{2C_3C_5R_3R_4g_m+2C_3C_5R_3+C_3C_5R_4}}(2C_3R_3g_m+C_3R_4g_m+2C_5R_4g_m+2C_5)}{2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_3R_4g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3+R_4}}+2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_3\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3+R_4}}+2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3+R_4}}}$

K-LP: $\frac{R_4}{2}$

K-HP: $-\frac{R_3R_4}{2R_3R_4g_m+2R_3+R_4}$

K-BP: $\frac{C_3R_3R_4g_m-C_5R_4}{2C_3R_3g_m+C_3R_4g_m+2C_5R_4g_m+2C_5}$

Qz: None

Wz: $\frac{\sqrt{-g_m}}{\sqrt{C_3}\sqrt{C_5}\sqrt{R_3}}$

10.5 X-INVALID-WZ-5 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_3C_5R_3R_4R_5s^2 + R_4R_5g_m - R_4 + s (C_3R_3R_4R_5g_m - C_3R_3R_4 - C_5R_4R_5)}{2R_4g_m + 2R_5g_m + s^2 (2C_3C_5R_3R_4R_5g_m + 2C_3C_5R_3R_5 + C_3C_5R_4R_5) + s (2C_3R_3R_4g_m + 2C_3R_3R_5g_m + 2C_3R_3 + C_3R_4R_5g_m + C_3R_4 + 2C_5R_4R_5g_m + 2C_5R_5) + 2}$$

Parameters:

Q: $\frac{2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_3R_4\sqrt{R_5g_m}\sqrt{\frac{R_4g_m}{2R_3R_4g_m+2R_3+R_4}}+\frac{R_5g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{1}{2R_3R_4g_m+2R_3+R_4}+2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_3\sqrt{R_5}\sqrt{\frac{R_4g_m}{2R_3R_4g_m+2R_3+R_4}}+\frac{R_5g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{1}{2R_3R_4g_m+2R_3+R_4}+\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_4\sqrt{R_5}\sqrt{\frac{R_4g_m}{2R_3R_4g_m+2R_3+R_4}}+\frac{R_5g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{1}{2R_3R_4g_m+2R_3+R_4}}{2C_3R_3R_4g_m+2C_3R_3R_5g_m+2C_3R_3+C_3R_4R_5g_m+C_3R_4+2C_5R_4R_5g_m+2C_5R_5}$

wo: $\sqrt{\frac{2R_4g_m+2R_5g_m+2}{2C_3C_5R_3R_4R_5g_m+2C_3C_5R_3R_5+C_3C_5R_4R_5}}$

bandwidth: $\frac{\sqrt{\frac{2R_4g_m+2R_5g_m+2}{2C_3C_5R_3R_4R_5g_m+2C_3C_5R_3R_5+C_3C_5R_4R_5}}(2C_3R_3R_4g_m+2C_3R_3R_5g_m+2C_3R_3+C_3R_4R_5g_m+C_3R_4+2C_5R_4R_5g_m+2C_5R_5)}{2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_3R_4\sqrt{R_5g_m}\sqrt{\frac{R_4g_m}{2R_3R_4g_m+2R_3+R_4}}+\frac{R_5g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{1}{2R_3R_4g_m+2R_3+R_4}+2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_3\sqrt{R_5}\sqrt{\frac{R_4g_m}{2R_3R_4g_m+2R_3+R_4}}+\frac{R_5g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{1}{2R_3R_4g_m+2R_3+R_4}+\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_4\sqrt{R_5}\sqrt{\frac{R_4g_m}{2R_3R_4g_m+2R_3+R_4}}+\frac{R_5g_m}{2R_3R_4g_m+2R_3+R_4}+\frac{1}{2R_3R_4g_m+2R_3+R_4}}}$

K-LP: $\frac{R_4R_5g_m-R_4}{2R_4g_m+2R_5g_m+2}$

K-HP: $-\frac{R_3R_4}{2R_3R_4g_m+2R_3+R_4}$

K-BP: $\frac{C_3R_3R_4R_5g_m-C_3R_3R_4-C_5R_4R_5}{2C_3R_3R_4g_m+2C_3R_3R_5g_m+2C_3R_3+C_3R_4R_5g_m+C_3R_4+2C_5R_4R_5g_m+2C_5R_5}$

Qz: None

Wz: $\frac{\sqrt{-R_5g_m+1}}{\sqrt{C_3}\sqrt{C_5}\sqrt{R_3}\sqrt{R_5}}$

10.6 X-INVALID-WZ-6 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, R_4, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_4 g_m + s^2 (C_3 C_5 R_3 R_4 R_5 g_m - C_3 C_5 R_3 R_4) + s (C_3 R_3 R_4 g_m + C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^2 (2C_3 C_5 R_3 R_4 g_m + 2C_3 C_5 R_3 R_5 g_m + 2C_3 C_5 R_3 + C_3 C_5 R_4 R_5 g_m + C_3 C_5 R_4) + s (2C_3 R_3 g_m + C_3 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_3R_4g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_3R_5g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_3\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_4R_5g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}}{2C_3R_3g_m+C_3R_4g_m+2C_5R_4g_m+2C_5R_5g_m+2C_5}$$

$$\text{wo: } \sqrt{2}\sqrt{\frac{g_m}{2C_3C_5R_3R_4g_m+2C_3C_5R_3R_5g_m+2C_3C_5R_3+C_3C_5R_4R_5g_m+C_3C_5R_4}}$$

$$\text{bandwidth: } \frac{\sqrt{2}\sqrt{\frac{g_m}{2C_3C_5R_3R_4g_m+2C_3C_5R_3R_5g_m+2C_3C_5R_3+C_3C_5R_4R_5g_m+C_3C_5R_4}}(2C_3R_3g_m+C_3R_4g_m+2C_5R_4g_m+2C_5R_5g_m+2C_5)}{2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_3R_4g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_3R_5g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+2\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_3\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_4R_5g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+\sqrt{2}\sqrt{C_3}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}}$$

$$\text{K-LP: } \frac{R_4}{2}$$

$$\text{K-HP: } \frac{R_3R_4R_5g_m-R_3R_4}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}$$

$$\text{K-BP: } \frac{C_3R_3R_4g_m+C_5R_4R_5g_m-C_5R_4}{2C_3R_3g_m+C_3R_4g_m+2C_5R_4g_m+2C_5R_5g_m+2C_5}$$

$$\text{Qz: None}$$

$$\text{Wz: } \sqrt{\frac{g_m}{C_3C_5R_3R_5g_m-C_3C_5R_3}}$$

10.7 X-INVALID-WZ-7 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_5 g_m + s^2 (C_3 C_4 R_3 R_4 R_5 g_m - C_3 C_4 R_3 R_4) + s (C_3 R_3 R_5 g_m - C_3 R_3 + C_4 R_4 R_5 g_m - C_4 R_4) - 1}{2g_m + s^2 (2C_3 C_4 R_3 R_4 g_m + 2C_3 C_4 R_3 R_5 g_m + 2C_3 C_4 R_3 + C_3 C_4 R_4 R_5 g_m + C_3 C_4 R_4) + s (2C_3 R_3 g_m + C_3 R_5 g_m + C_3 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4)}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_3R_4g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+2\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_3R_5g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+2\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_3\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_4R_5g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_4\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}}{2C_3R_3g_m+C_3R_5g_m+C_3+2C_4R_4g_m+2C_4R_5g_m+2C_4}$$

$$\text{wo: } \sqrt{2}\sqrt{\frac{g_m}{2C_3C_4R_3R_4g_m+2C_3C_4R_3R_5g_m+2C_3C_4R_3+C_3C_4R_4R_5g_m+C_3C_4R_4}}$$

$$\text{bandwidth: } \frac{\sqrt{2}\sqrt{\frac{g_m}{2C_3C_4R_3R_4g_m+2C_3C_4R_3R_5g_m+2C_3C_4R_3+C_3C_4R_4R_5g_m+C_3C_4R_4}}(2C_3R_3g_m+C_3R_5g_m+C_3+2C_4R_4g_m+2C_4R_5g_m+2C_4)}{2\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_3R_4g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+2\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_3R_5g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+2\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_3\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_4R_5g_m\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}+\sqrt{2}\sqrt{C_3}\sqrt{C_4}R_4\sqrt{\frac{g_m}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}}}$$

$$\text{K-LP: } \frac{R_5g_m-1}{2g_m}$$

$$\text{K-HP: } \frac{R_3R_4R_5g_m-R_3R_4}{2R_3R_4g_m+2R_3R_5g_m+2R_3+R_4R_5g_m+R_4}$$

$$\text{K-BP: } \frac{C_3R_3R_5g_m-C_3R_3+C_4R_4R_5g_m-C_4R_4}{2C_3R_3g_m+C_3R_5g_m+C_3+2C_4R_4g_m+2C_4R_5g_m+2C_4}$$

$$\text{Qz: None}$$

$$\text{Wz: } \frac{1}{\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}$$

11 X-PolynomialError